C-STGB Real-Time Financial Surveillance Architecture & Conformal Triage Pipeline

1. Streaming Ingest
SWIFT / ISO-20022 / Kafka
Apache Arrow Buffers
Time-Causal Ingestion
DuckDB Micro-Batches
t_e <= t_eval (No Leak)
12-D Canonical Invariants:
Phi_flow, Hawkes lambda_u,
s_fwd, s_bwd, [mu_A, sigma_A^2]

Single-event: 0.11 ms

2. C-STGB Neural Core
3-Band Harmonic Temporal GNN
+ Context Edge Gate g_ij
Prunes Chaff Camouflage
(delta_floor >= 0.10, 92.1% Ret.)
Latent GraphSMOTE C_k
Evidence Adaptive Fusion alpha_u
89.60% Cold-Start Minority F1
Cost-Sensitive Bayes Threshold:
tau* = 0.18 (15:1 Cost Ratio)

Inference: 0.16 ms

Class-Cond. CRC Triage
$1 - \alpha_i(\hat{q}^{(0)}, \hat{q}^{(1)}) \geq 99\%$

$\Gamma = \{1\}$ Tier 1: High-Risk Quarantine
Immediate Quarantine Hold | Assisted FinCEN SAR Dossier
18 alerts / 10,000 tx (0.18%) | SHA-256 Merkle Audit Receipt

$\Gamma = \{0, 1\}$ Tier 2: Boundary Review Desk
Abstention Set: Escalate to Senior Compliance Examiner
50 alerts / 10,000 tx (0.50%) | 4-Eyes AML Review Desk

$\Gamma = \{0\}$ Tier 3: Straight-Through Clear
Automated Wire-Speed Release (STP) | Residual Leak <0.12%
9,932 tx / 10,000 tx (99.32%) | Zero Latency Impact

MATHEMATICAL FORMULATIONS, RISK BOUNDS & REAL-TIME SYSTEM PERFORMANCE SLAS

Tri-Band Temporal: $\Phi_{\text{time}}(\Delta t) = [\cos(\omega_k \Delta t), \sin(\omega_k \Delta t)]_{k=1}^{3}$

Edge Trust Filter: $\hat{g}_{ij} = \sigma(\mathbf{w}_g^\top [\mathbf{h}_i \,\|\, \mathbf{h}_j \,\|\, \mathbf{e}_{ij}]), \quad \mathbf{A}_{\text{pruned}} = \mathbf{A} \odot \mathbf{I}(\hat{g}_{ij} \geq \delta_{\text{floor}})$

12-D Flow Dynamics: $\tilde{\Phi}_{\text{flow}} = \ln(1 + \sum A_{\text{out}}/(\sum A_{\text{in}} + \epsilon)), \quad \lambda_u(t) = \mu_u + \sum_{t_i < t} \alpha e^{-\beta(t - t_i)}$

Class-Cond. CRC: $\hat{q}^{(y)} = \text{Quantile}_{1-\alpha_i}(\{s_i^{(y)}\}), \quad \mathbf{P}(Y \in \Gamma(X) \mid Y = y) \geq 1 - \alpha$

Triage Dispatch: $\Gamma(u) = \{y : s_u^{(y)} \leq \hat{q}^{(y)}\}, \quad \mathbf{E}[|\Gamma|] = 1.0050, \quad \mathbf{P}(\Gamma = \varnothing) = 0.00\%$

Fast-Path Latency: $0.45\,\text{ms} < 10\,\text{ms}$ **SLA | Queue:** -99.50% **Workload Cut | Peak VRAM:** 428 MB